SRIRAM REPAKA

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TECHNICAL PROFILE

- **Firmware & Embedded C:** C, Python, Embedded Linux, RTOS, UART, CAN Bus, SPI, I2C, SystemVerilog
- **Robotics & Vision:** ROS/ROS2, PyTorch, OpenCV, SegNet, DetectNet, LiDAR Processing, ArUco Tracking
- **Hardware & Tools:** ESP32, Jetson Nano, Raspberry Pi, Fusion 360 (CAD/3D Printing), JIRA, Git, LTSpice, Proteus
- **Languages:** English (C1 - Professional), German (A2 - Intermediate), Telugu (Native)

EDUCATION

M.Sc. in Electronics Engineering (Microsystems) 10/2024 – Present
Hochschule Bremen, Germany | Current Grade: 1.7

- **Focus:** AI in Robotics, Advanced Hardware Verification, Image Processing, Mixed-Signal IC Design, MEMS

B.Tech. in Electrical and Electronics Engineering 08/2018 – 08/2022
Mahindra École Centrale, India | Grade: 2.0 (German equivalent)

- **Focus:** Microprocessors, Computer Architecture, Embedded Systems, IoT

PROFESSIONAL EXPERIENCE

Embedded Software Developer | Associate II 08/2022 – 03/2024
Capgemini Engineering (Client: GoPro) — Bangalore, India

- Developed and modified production application-layer (APPC) camera firmware features in C.
- Conducted Root Cause Analysis (RCA) for complex system bugs by evaluating real-time UART logs via Tera Term.
- Utilized hardware debug boards and logic analyzers to trace software execution and hardware interactions.
- Managed bug life-cycles in JIRA, maintained version control in Git, and conducted code reviews with lead Architect.

R&D Embedded Systems Intern 08/2020 – 10/2020
Cellerite Systems — Hyderabad, India

- Prototyped an EV charging communication and current/voltage monitoring system using PIC18F458.
- Implemented CAN bus communication protocol to enable real-time device-to-station data exchange.
- Simulated microcontroller hardware and CAN protocol execution using Proteus and MPLAB IDE.

SELECTED HARDWARE & ROBOTICS PROJECTS

Smart Desktop Assistant & Custom Hardware Enclosure 06/2026 – Present

- Designed an ESP32-driven desktop companion featuring task logging, Pomodoro timer, and Wi-Fi sync.
- Modeled a custom enclosure with precise speaker/battery mounts in Fusion 360; 3D-printed final chassis.

Analog IC Filter Design: 4th-Order Linkwitz-Riley *(Hochschule Bremen)* 03/2026 – 07/2026

- Sized and simulated a 2-stage 5T-OTA using 130nm open-source SG13G2 process in Xschem and ngspice.
- Performed comprehensive Process, Voltage, Temperature (PVT) parameter and stability analysis.

AI-Powered Autonomous JetBot *(Hochschule Bremen)* 10/2025 – 01/2026

- Built an autonomous lane-following robot with obstacle avoidance using Jetson Nano, LiDAR, and IMU.
- Deployed real-time SegNet and DetectNet models for onboard inference and dynamic boundary detection.

Autonomous BRIO Maze Automation *(Hochschule Bremen)* 10/2024 – 01/2025

- Automated a tilting physical maze game using real-time ArUco marker pose estimation in OpenCV.
- Developed a closed-loop Python control algorithm driving Dynamixel servos for precision ball navigation.

Assistive LiDAR Navigation Aid *(B.Tech Thesis)* 08/2021 – 05/2022

- Engineered a walking stick for the visually impaired integrating 3D LiDAR Time-of-Flight depth mapping.
- Designed a custom pneumatic shock absorber mechanism to stabilize camera footage on rough terrain.